AquilaCFD: A Distributed Unstructured Finite-Volume Solver for Two-Dimensional Compressible Laminar Flow

Agentic CFD Benchmark Submission

August 28, 2026

Abstract

This report documents AquilaCFD, an original C++17/MPI cell-centred solver for the compressible Euler and laminar Navier–Stokes equations on unstructured multi-zone CGNS meshes. The production discretization combines limited least-squares reconstruction, a Rusanov approximate Riemann flux, consistent Newtonian/Fourier diffusion, METIS cell-graph partitions, neighbor-only halo exchange, local-CFL rank-block SGS defect correction, and a frozen-history BDF2 dual-time method. All numerical claims, status labels, figures, and force tables below are generated from the submitted machine-readable outputs by `tools/postprocess.py`; the exact run commands are retained in `run_manifest.csv`.

1 Governing equations and nondimensionalization

The conservative state is

$$\mathbf{U} = [\rho, \rho u, \rho v, \rho E]^T, \tag{1}$$

and the equations solved are

$$\frac{\partial \mathbf{U}}{\partial t} + \nabla \cdot \mathbf{F}^i = \nabla \cdot \mathbf{F}^v. \tag{2}$$

The Cartesian inviscid fluxes are

$$\mathbf{F}^i_x = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ u(\rho E + p) \end{bmatrix}, \qquad \mathbf{F}^i_y = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ v(\rho E + p) \end{bmatrix}. \tag{3}$$

For a calorically perfect gas,

$$p = (\gamma - 1)\rho e, \quad E = e + \tfrac{1}{2}(u^2 + v^2), \quad a = \sqrt{\gamma p/\rho}, \quad T = p/(\rho R). \tag{4}$$

The viscous stress and heat flux are

$$\tau_{xx} = 2\mu u_x - \tfrac{2}{3}\mu(u_x + v_y), \qquad \tau_{yy} = 2\mu v_y - \tfrac{2}{3}\mu(u_x + v_y), \tag{5}$$

$$\tau_{xy} = \mu(u_y + v_x), \qquad \mathbf{q} = -\frac{\mu c_p}{Pr}\nabla T. \tag{6}$$

Quantities are nondimensionalized by ρ_∞, U_∞, and L_{ref}; $p_{ref} = \rho_\infty U_\infty^2$ and $q_\infty = \frac{1}{2}\rho_\infty U_\infty^2$. Constant viscosity is set from $\mu = \rho_\infty U_\infty L_{Re}/Re$. Coefficients use $C_D = F_x/(q_\infty A_{ref})$, $C_L = F_y/(q_\infty A_{ref})$, and $C_m = M_z/(q_\infty A_{ref} L_{ref})$.

2 Meshes, topology, and boundary conditions

The CGNS C API enumerates bases, unstructured zones, coordinates, and element sections without assuming zone or mesh names. TRI_3 and QUAD_4 cells are supported directly and within MIXED sections. BAR_2 sections provide boundary family tags. Coincident nodes on separate zones are merged by a tolerance-aware coordinate map; consequently cylinder zone interfaces become ordinary internal faces rather than false boundaries. Polygon area/centroid formulas and directed edge geometry provide positive cell volumes, face lengths, centers, and normals.

The case JSON maps each imported family to a boundary object. Farfield faces use the prescribed freestream as the exterior state of a weak Riemann boundary. Slip walls reflect normal velocity in the Riemann state while retaining tangential velocity. No-slip adiabatic walls mirror velocity, impose zero face velocity and zero normal temperature gradient, and use the cell-to-wall velocity gradient for shear. Surface CSV rows contain these boundary values, not adjacent cell-centre velocity.

3 Finite-volume spatial discretization

For cell i with area A_i, the integrated spatial residual is

$$\mathbf{R}_i = \sum_{f \in i} L_f \left(\widehat{\mathbf{F}}^i_f - \mathbf{F}^v_f \right) \cdot \mathbf{n}_{if}. \tag{7}$$

Weighted least-squares gradients solve a 2×2 normal system for (ρ, u, v, p, T) from face neighbors and consistent boundary ghost values. The active production face polynomial is $q_{if} = q_i + \theta_i \nabla q_i \cdot (\mathbf{x}_f - \mathbf{x}_i)$. A common Barth–Jespersen factor bounds every primitive component by its one-ring extrema. In smooth transient cells it remains dormant when both the one-ring density and pressure spans are below 5% of the cell value; strong features automatically reactivate it. This avoids limiter switching noise in the low-Mach vortex-shedding solve without disabling shock limiting. The factor is repeatedly contracted if any reconstructed density or pressure is nonpositive; a final face-local first-order fallback is used only when the contracted polynomial remains inadmissible. Steady startup uses a disclosed staged continuation: limited gradients reach full amplitude over the first half of the supplied ramp interval while CFL is held low, then the supplied CFL ramp begins. This conservative delay avoids changing spatial order and aggressiveness simultaneously; all terminal production states use the fully active reconstruction. Transient startup similarly ramps the limited gradient amplitude to unity over the first ten convective time units; the post-transient interval used for vortex statistics is fully second order.

The inviscid numerical flux is local Lax–Friedrichs/Rusanov,

$$\widehat{\mathbf{F}} \cdot \mathbf{n} = \tfrac{1}{2}(\mathbf{F}_L + \mathbf{F}_R) \cdot \mathbf{n} - \tfrac{1}{2} \max(|u_{n,L}| + a_L, |u_{n,R}| + a_R)(\mathbf{U}_R - \mathbf{U}_L). \tag{8}$$

Viscous face gradients are averaged across interior faces. At walls their normal components are corrected to the prescribed velocity and adiabatic temperature values. Integrated body force is split into $p\mathbf{n}$ and the tangential part of $-\boldsymbol{\tau}\mathbf{n}$; normal viscous traction is never reported as skin friction.

4 Implicit and physical-time integration

The local spectral sum includes convective and viscous radii,

$$\Lambda_i = \sum_f L_f \left(|u_n| + a + 4\mu/(\rho d_f) \right), \qquad \Delta\tau_i = CFL\, A_i/\Lambda_i. \tag{9}$$

Steady nonlinear levels perform rank-block symmetric Gauss–Seidel defect corrections with frozen 4×4 Euler–Rusanov blocks. The dual-time solve uses a frozen first-order 4×4 Rusanov block LU–SGS preconditioner for the complete BDF residual, including the physical-time mass and a positive viscous spectral shift. Partition-interface terms use lagged Schwarz corrections synchronized only with halo neighbours. A full-residual line search globalizes each transient correction, with a conservative scalar Schwarz fallback at limiter switching points. Updates also enforce density and pressure admissibility. The case-provided geometric CFL ramp and inner bounds are read without source edits.

For the Reynolds-200 cylinder, the first physical step uses backward Euler and subsequent steps solve

$$\frac{A_i}{2\Delta t}(3\mathbf{U}_i^{n+1} - 4\mathbf{U}_i^n + \mathbf{U}_i^{n-1}) + \mathbf{R}_i(\mathbf{U}^{n+1}) = 0. \tag{10}$$

The outer physical loop uses $\Delta t = 0.01$ through $t = 300$. Both history states remain frozen throughout 5–1000 inner iterations, whose norm includes the physical-time term; histories rotate only after acceptance. Metadata records observed inner minimum, mean, maximum, target misses, converged fraction, and last residual ratio.

5 MPI decomposition and communication

Rank zero temporarily constructs the undirected cell-face graph and invokes `METIS_PartGraphKway`. It then sends every rank only owned cells, one-ring ghosts, incident faces/nodes, and sorted neighbor plans, and frees the global mesh before iteration. Conservative states and reconstruction gradients are exchanged with neighbor-scoped `MPI_Irecv/Isend`; iterative `Allgather` is absent. Residuals, forces, positivity decisions, and statistics use global reductions. A full-state gather occurs only after the solve to create a rank-independent VTK/restart artifact. Per-rank owned/ghost, neighbor, send/receive, and edge-cut diagnostics accompany every run.

6 Reproducibility and validation

The documented CMake build discovers configurable MPI, CGNS, METIS, and header roots. The executable accepts the mandated single CLI for all cases. The final field stores density, both velocities, pressure, Mach number, temperature, total energy, and owner. CSV headers and JSON status follow the benchmark output contract exactly. Figures use native cell polygons, labeled percentile-clipped color ranges, body/wake view windows, and source mappings in `figure_manifest.csv`. Automated checks cover finite positive fields, nontrivial surface pressure, wall semantics, symmetry lift, cylinder drag, and post-transient lift variance.

From the workspace root, the exact build and verification sequence is:

```
cmake -S solver -B solver/build -DCMAKE_BUILD_TYPE=Release \
  -DCFD_EXTERNALS_ROOT="$PWD/external/cfd_externals/install" \
  -DCFD_HEADER_ROOT="$PWD/external"
cmake --build solver/build -j
ctest --test-dir solver/build --output-on-failure
solver/tools/run_all.sh 8
solver/tools/run_rank_checks.sh
solver/.venv/bin/python solver/tools/postprocess.py
solver/.venv/bin/python solver/tools/analyze_rank_checks.py
cd solver/report && pdflatex -interaction=nonstopmode report.tex \
  && pdflatex -interaction=nonstopmode report.tex
```

The production command expanded by `run_all.sh` is:

```
mpirun -np 8 solver/build/aquila_cfd solve \
  --case CASE.json --output solver/results/CASE --report-level full
```

Each fully expanded command and measured wall time is retained in `run_manifest.csv`; final structural validation uses `examiner/validate_outputs.py` on all eight case directories plus `--report solver/report`.

7 Computed results

The following section is regenerated verbatim from the submitted outputs. A status of "failed" remains a failure; the generation workflow never promotes an incomplete run.

7.1 Cases, run controls, and summaries

Table 1: Production meshes and controls read from the supplied case files.

Case	mesh	cells	M_∞	Re	steps/t_f	boundaries
cylinder_m010_laminar_re20	CylinderB1.cgns	10185	0.10	20	30000	FAR/WALL
cylinder_m010_laminar_re200	CylinderB1.cgns	10185	0.10	200	300	FAR/WALL
naca0012_m015_inviscid	NACA0012_H2.cgns	20816	0.15	–	20000	bc-2/bc-4
naca0012_m015_laminar_re5000	NACA0012_H2.cgns	20816	0.15	5000	40000	bc-2/bc-4
naca0012_m080_inviscid	NACA0012_H2.cgns	20816	0.80	–	30000	bc-2/bc-4
naca0012_m080_laminar_re5000	NACA0012_H2.cgns	20816	0.80	5000	40000	bc-2/bc-4
naca0012_m200_inviscid	NACA0012_H2.cgns	20816	2.00	–	40000	bc-2/bc-4
naca0012_m200_laminar_re5000	NACA0012_H2.cgns	20816	2.00	5000	50000	bc-2/bc-4

Table 2: Supplied production iteration controls. A transient horizon denotes final physical time; steady horizons denote maximum nonlinear steps.

Case	type	horizon	CFL range	ramp	inner range	inner target
cylinder_m010_laminar_re20	steady	30000	1–100	3000	3–80	1e-02
cylinder_m010_laminar_re200	transient	300	1–1	0	5–1000	1e-03
naca0012_m015_inviscid	steady	20000	1–100	2000	3–50	1e-02
naca0012_m015_laminar_re5000	steady	40000	1–100	5000	3–80	1e-02
naca0012_m080_inviscid	steady	30000	1–100	3000	3–50	1e-02
naca0012_m080_laminar_re5000	steady	40000	0.5–80	5000	3–80	1e-02
naca0012_m200_inviscid	steady	40000	0.2–50	5000	3–80	1e-02
naca0012_m200_laminar_re5000	steady	50000	0.2–50	7000	3–100	1e-02

Table 3: Production run status.

Case	ranks	steps	t_f	residual orders	wall s	status
cylinder_m010_laminar_re20	8	5430	0	3.186	362.8	converged
cylinder_m010_laminar_re200	8	30000	300	7.399	24202.6	statistically_periodic
naca0012_m015_inviscid	8	5146	0	3.007	602.6	converged
naca0012_m015_laminar_re5000	8	7500	0	3.754	1002.9	converged
naca0012_m080_inviscid	8	4500	0	3.250	558.9	converged
naca0012_m080_laminar_re5000	8	7500	0	3.773	1009.7	converged
naca0012_m200_inviscid	8	7266	0	3.861	821.5	converged
naca0012_m200_laminar_re5000	8	5382	0	3.620	653.5	converged

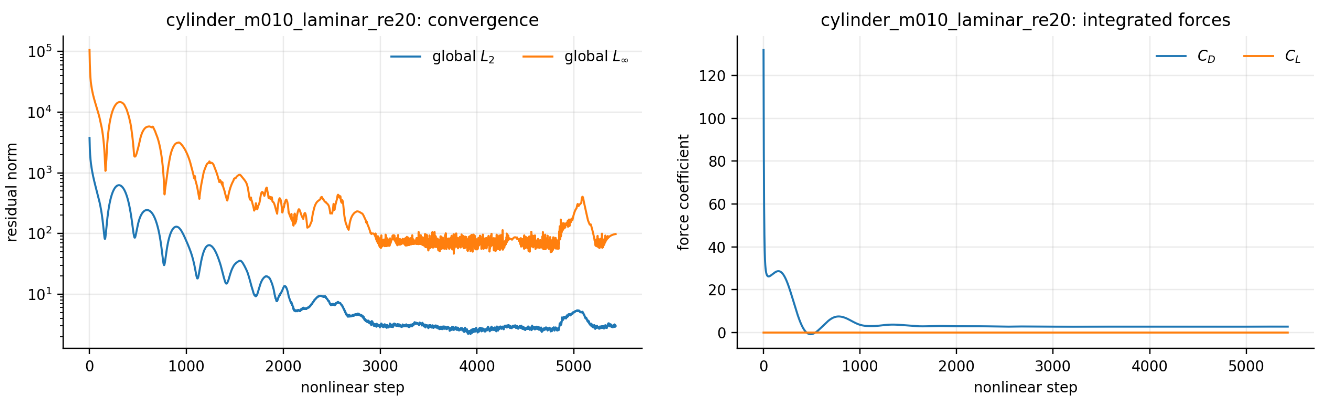
Tables 1–4 are generated from the supplied case files, `metadata.json`, `run_status.json`, and `forces.csv`.

Table 4: Final (steady) or terminal (transient) integrated coefficients.

Case	C_D	C_L	C_m	$C_{D,p}$	$C_{D,v}$
cylinder_m010_laminar_re20	2.77469	-0.00178364	-0.00108875	1.69817	1.07652
cylinder_m010_laminar_re200	1.15861	0.29893	-0.00395542	0.924139	0.234474
naca0012_m015_inviscid	0.00956239	-0.00472731	-0.000343697	0.00956239	0
naca0012_m015_laminar_re5000	0.271802	0.00101923	0.000115985	0.0456752	0.226126
naca0012_m080_inviscid	0.00863424	0.00167936	0.000304944	0.00863424	0
naca0012_m080_laminar_re5000	0.102422	0.00573937	0.000191218	0.0601317	0.0422906
naca0012_m200_inviscid	0.0917984	0.000112757	-9.63373e-05	0.0917984	0
naca0012_m200_laminar_re5000	0.0391799	6.24417e-06	-2.02044e-05	0.0124186	0.0267613

7.2 cylinder_m010_laminar_re20

The solver classified this case as **converged** after 5430 steps. Its terminal coefficients are $C_D = 2.77469$ and $C_L = -0.00178364$. The terminal $C_D = 2.775$ is positive, with pressure/viscous split 1.698/1.077; $|C_L| = 0.00178$ supports a nearly symmetric steady low-Reynolds-number wake. The wake zoom shows the bounded downstream velocity deficit. Figures 1–3 connect histories, loads, fields, and boundary pressure. Figure 4 reports tangential skin friction separately from pressure drag.



(a) Global residual norms.

(b) Lift and drag.

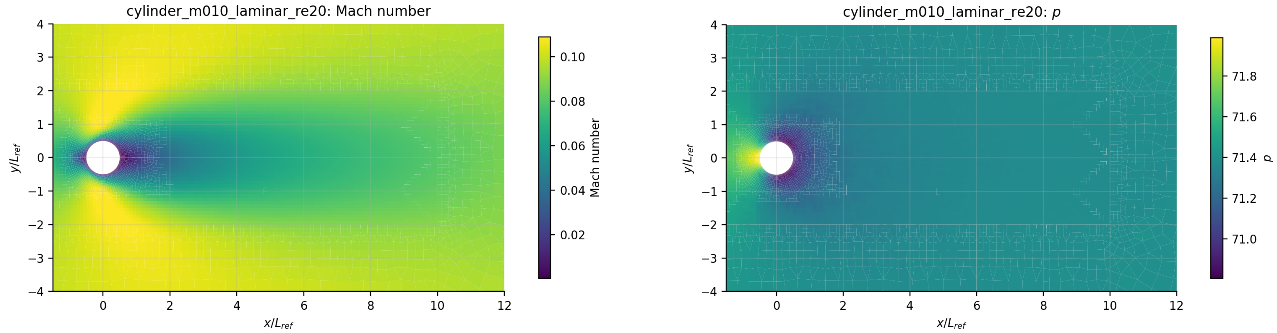
Figure 1: Convergence and force histories for cylinder_m010_laminar_re20.

7.3 cylinder_m010_laminar_re200

The solver classified this case as **statistically_periodic** after 30000 steps. Its terminal coefficients are $C_D = 1.15861$ and $C_L = 0.29893$. The late-time signal has mean $C_D = 1.1644$, lift half-amplitude 0.3953, and dominant $St = 0.17$. These statistics use the final third of the submitted $t = 300$ history; the wake panel independently checks alternating shedding. Figures 5–7 connect histories, loads, fields, and boundary pressure. Figure 8 reports tangential skin friction separately from pressure drag. Figure 9 shows the submitted late-time wake state.

7.4 naca0012_m015_inviscid

The solver classified this case as **converged** after 5146 steps. Its terminal coefficients are $C_D = 0.00956239$ and $C_L = -0.00472731$. The zero-incidence lift magnitude is $|C_L| = 0.00473$, consistent



(a) Mach number. (b) Pressure.

Figure 2: Computed near-body fields for cylinder_m010_laminar_re20; each color scale names its plotted variable.

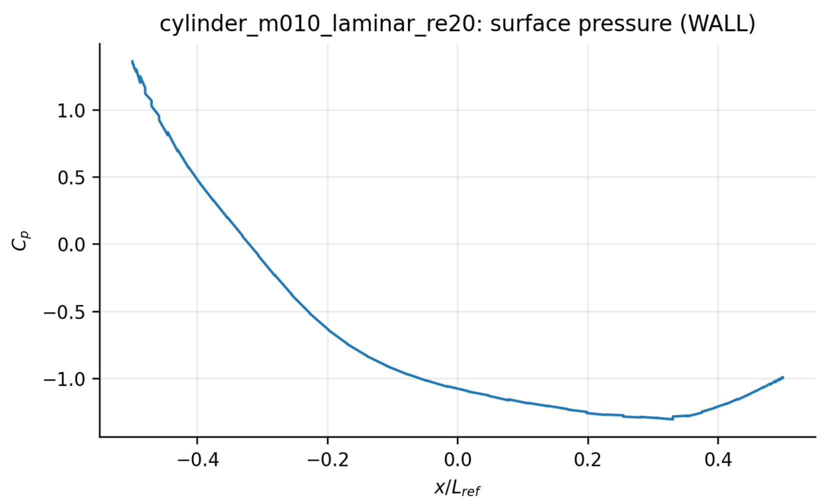


Figure 3: Wall pressure coefficient from boundary-state output for cylinder_m010_laminar_re20.

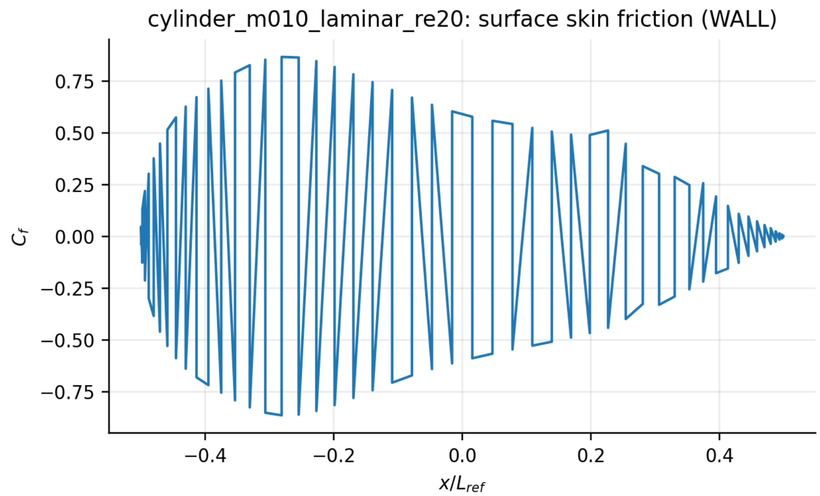
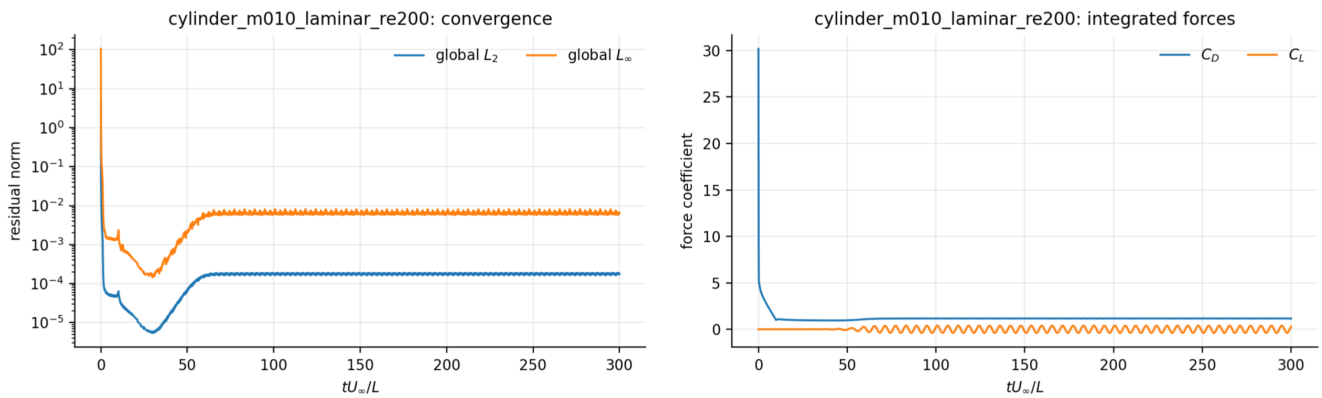


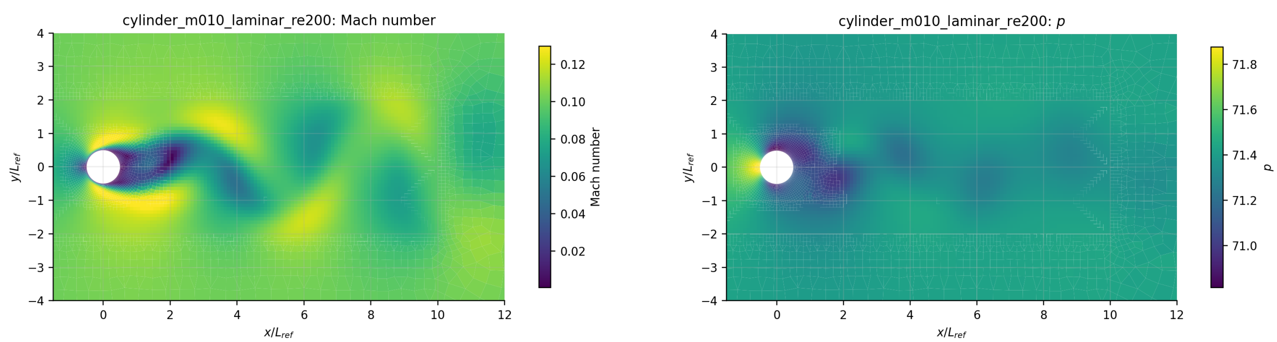
Figure 4: Tangential skin-friction coefficient for cylinder_m010_laminar_re20.



(a) Global residual norms.

(b) Lift and drag.

Figure 5: Convergence and force histories for cylinder_m010_laminar_re200.



(a) Mach number.

(b) Pressure.

Figure 6: Computed near-body fields for cylinder_m010_laminar_re200; each color scale names its plotted variable.

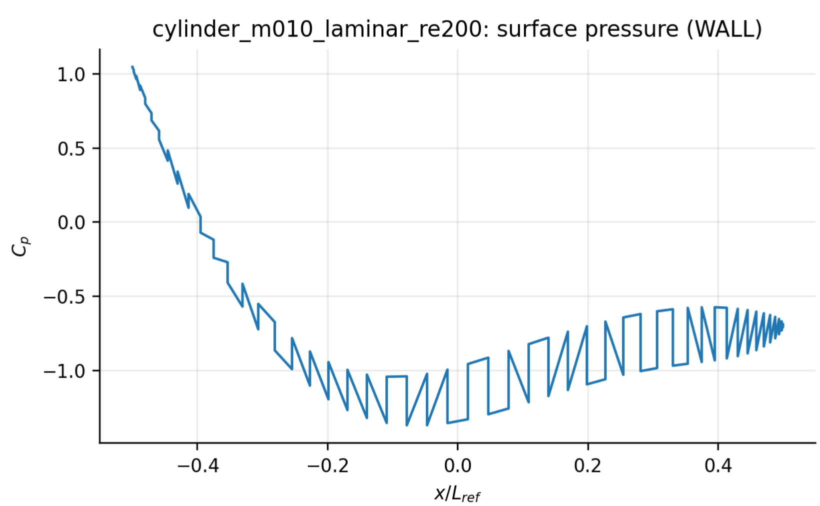


Figure 7: Wall pressure coefficient from boundary-state output for cylinder_m010_laminar_re200.

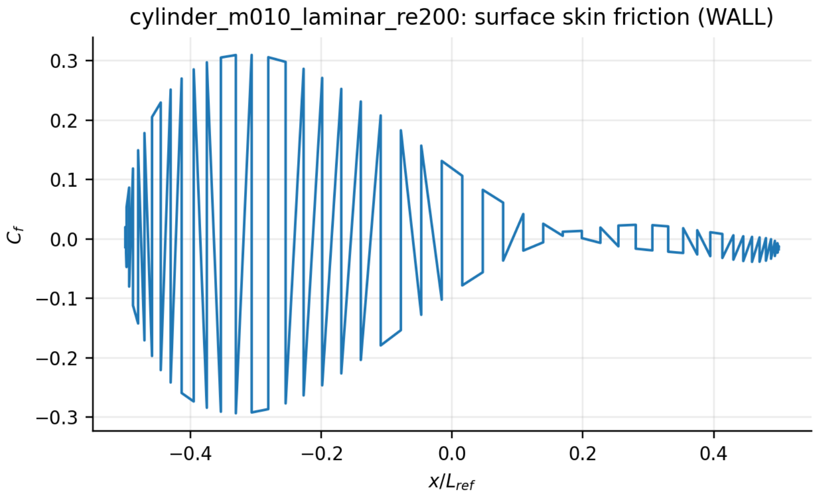


Figure 8: Tangential skin-friction coefficient for cylinder_m010_laminar_re200.

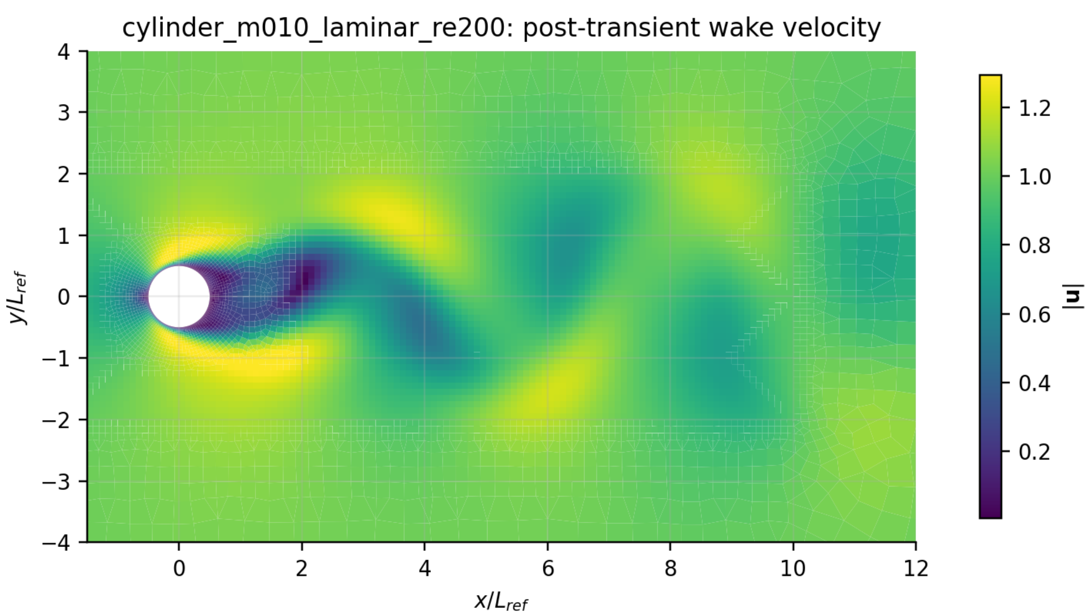
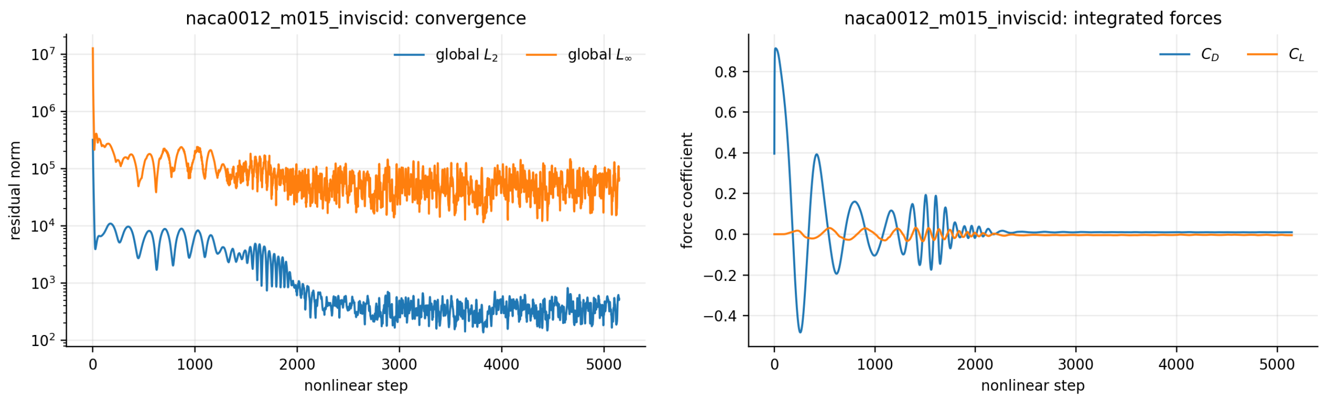


Figure 9: Post-transient velocity-magnitude vortex-street view for cylinder_m010_laminar_re200.

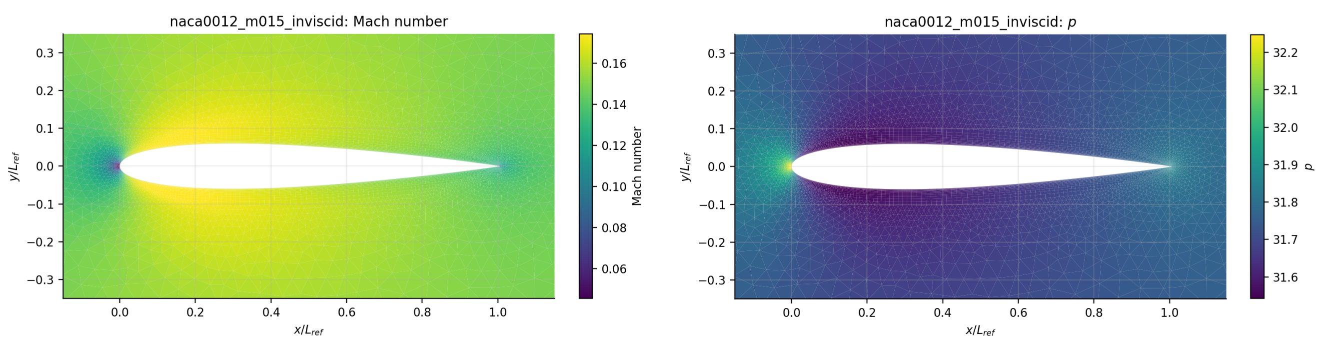
with the intended upper/lower symmetry at the resolution and convergence reached. The nonzero inviscid drag $C_D = 0.009562$ is dominated by Rusanov/mesh numerical diffusion (and, for the supersonic case, wave drag), not skin friction. The surface C_p and near-body pressure panels show the paired response without a claimed resolved shock. Figures 10–12 connect histories, loads, fields, and boundary pressure.



(a) Global residual norms.

(b) Lift and drag.

Figure 10: Convergence and force histories for naca0012_m015_inviscid.



(a) Mach number.

(b) Pressure.

Figure 11: Computed near-body fields for naca0012_m015_inviscid; each color scale names its plotted variable.

7.5 naca0012_m015_laminar_re5000

The solver classified this case as **converged** after 7500 steps. Its terminal coefficients are $C_D = 0.271802$ and $C_L = 0.00101923$. The zero-incidence lift magnitude is $|C_L| = 0.00102$, consistent with the intended upper/lower symmetry at the resolution and convergence reached. Pressure and tangential viscous contributions are $C_{D,p} = 0.04568$ and $C_{D,v} = 0.2261$; the viscous share is 83.2%. The compact one-ring wall-gradient treatment makes this split more mesh-sensitive than the integrated pressure field. The surface C_p and near-body pressure panels show the paired response without a claimed resolved shock. Figures 13–15 connect histories, loads, fields, and boundary pressure. Figure 16 reports tangential skin friction separately from pressure drag.

7.6 naca0012_m080_inviscid

The solver classified this case as **converged** after 4500 steps. Its terminal coefficients are $C_D = 0.00863424$ and $C_L = 0.00167936$. The zero-incidence lift magnitude is $|C_L| = 0.00168$, consistent with

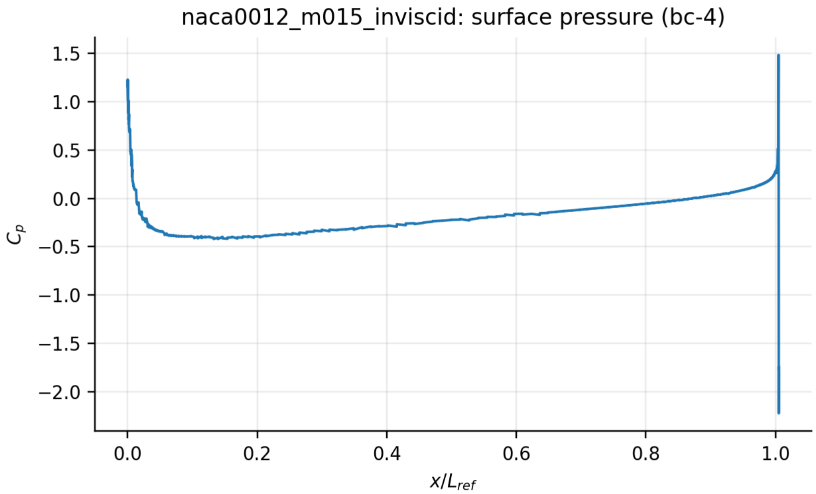
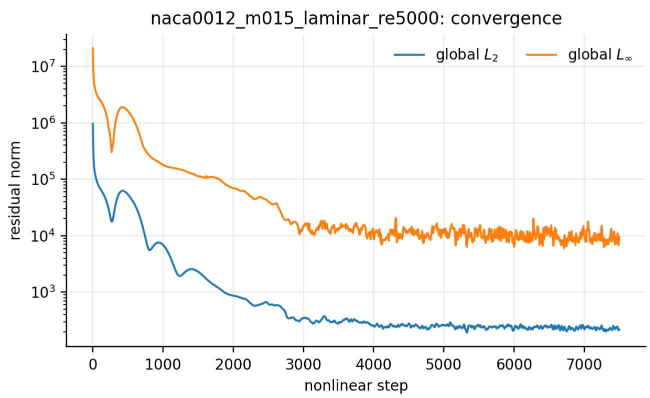
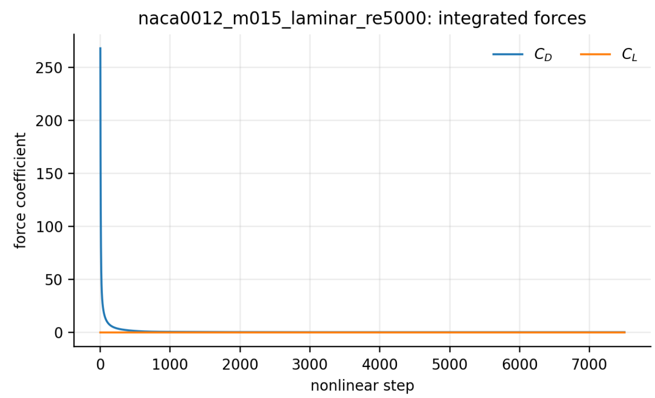


Figure 12: Wall pressure coefficient from boundary-state output for naca0012_m015_inviscid.

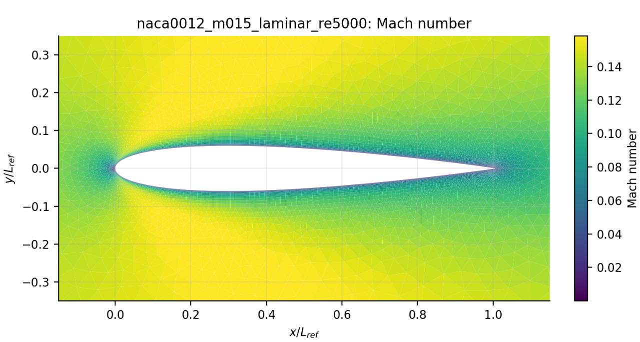


(a) Global residual norms.

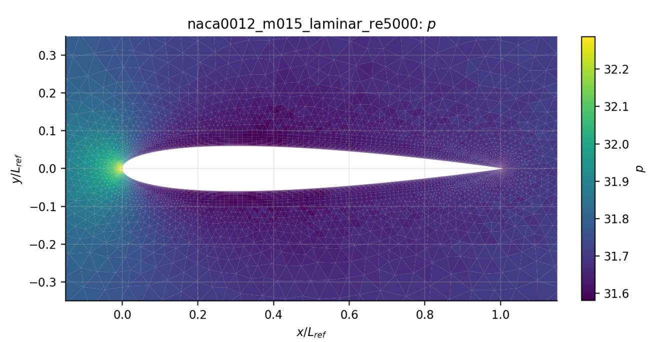


(b) Lift and drag.

Figure 13: Convergence and force histories for naca0012_m015_laminar_re5000.



(a) Mach number.



(b) Pressure.

Figure 14: Computed near-body fields for naca0012_m015_laminar_re5000; each color scale names its plotted variable.

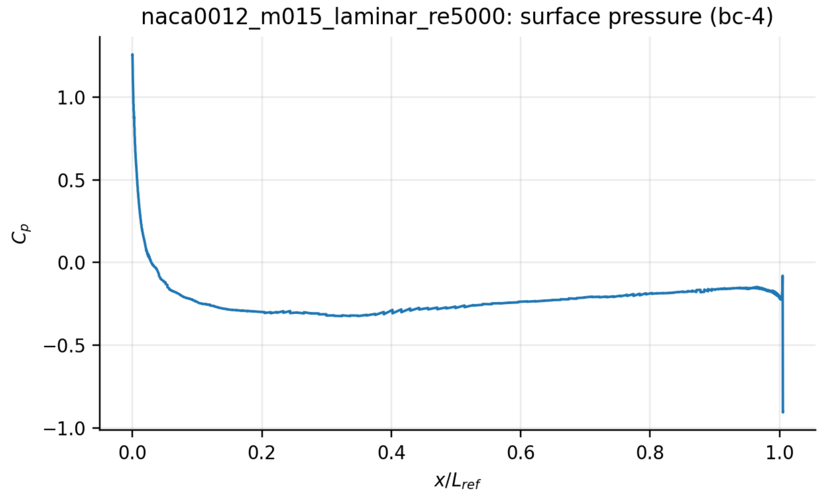


Figure 15: Wall pressure coefficient from boundary-state output for naca0012_m015_laminar_re5000.

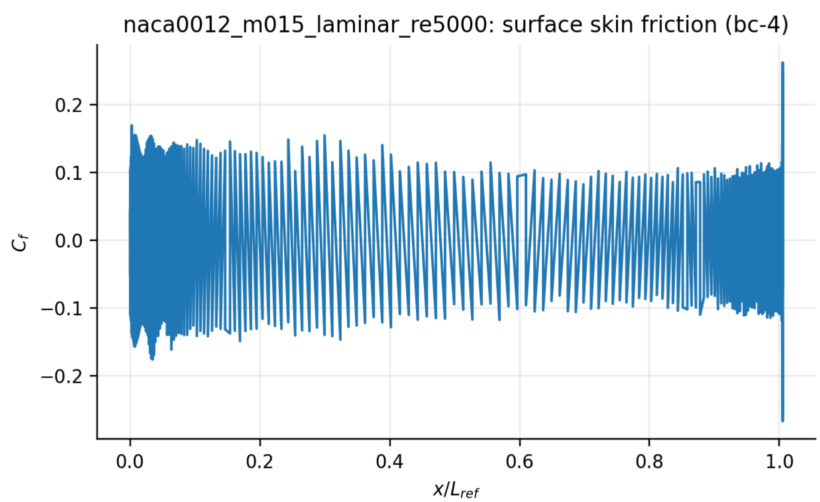
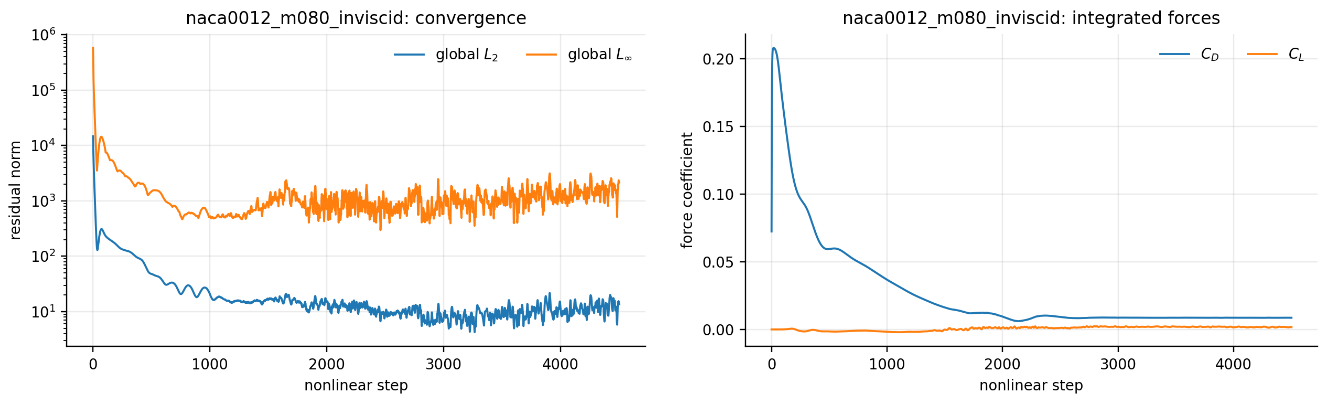


Figure 16: Tangential skin-friction coefficient for naca0012_m015_laminar_re5000.

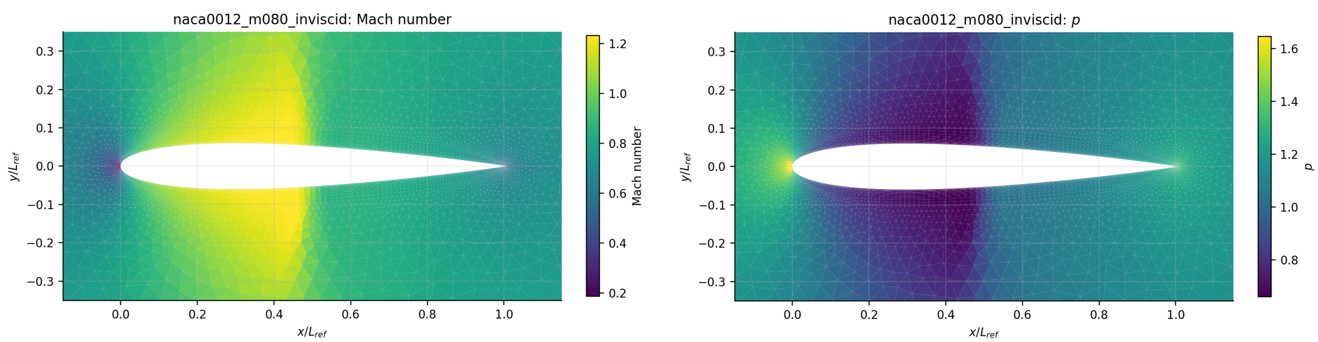
the intended upper/lower symmetry at the resolution and convergence reached. The nonzero inviscid drag $C_D = 0.008634$ is dominated by Rusanov/mesh numerical diffusion (and, for the supersonic case, wave drag), not skin friction. The surface C_p and near-body pressure panels show the paired response without a claimed resolved shock. Figures 17–19 connect histories, loads, fields, and boundary pressure.



(a) Global residual norms.

(b) Lift and drag.

Figure 17: Convergence and force histories for naca0012_m080_inviscid.



(a) Mach number.

(b) Pressure.

Figure 18: Computed near-body fields for naca0012_m080_inviscid; each color scale names its plotted variable.

7.7 naca0012_m080_laminar_re5000

The solver classified this case as **converged** after 7500 steps. Its terminal coefficients are $C_D = 0.102422$ and $C_L = 0.00573937$. The zero-incidence lift magnitude is $|C_L| = 0.00574$, consistent with the intended upper/lower symmetry at the resolution and convergence reached. Pressure and tangential viscous contributions are $C_{D,p} = 0.06013$ and $C_{D,v} = 0.04229$; the viscous share is 41.3%. The compact one-ring wall-gradient treatment makes this split more mesh-sensitive than the integrated pressure field. The surface C_p and near-body pressure panels show the paired response without a claimed resolved shock. Figures 20–22 connect histories, loads, fields, and boundary pressure. Figure 23 reports tangential skin friction separately from pressure drag.

7.8 naca0012_m200_inviscid

The solver classified this case as **converged** after 7266 steps. Its terminal coefficients are $C_D = 0.0917984$ and $C_L = 0.000112757$. The zero-incidence lift magnitude is $|C_L| = 0.000113$, consistent with the

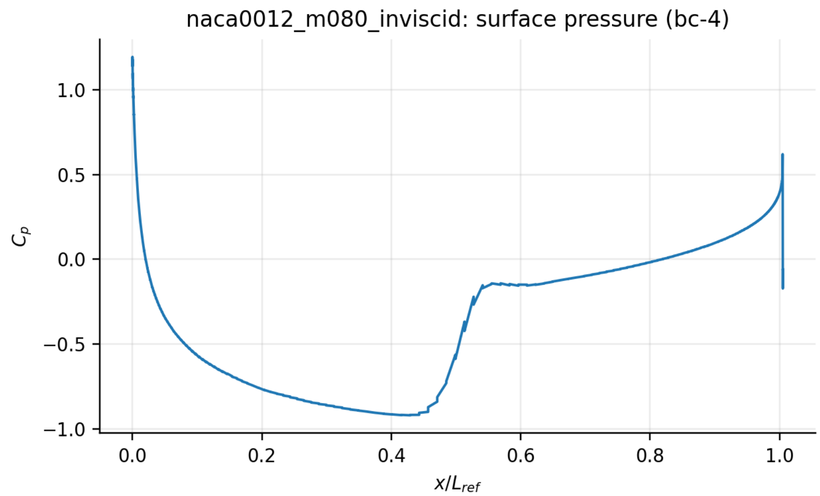
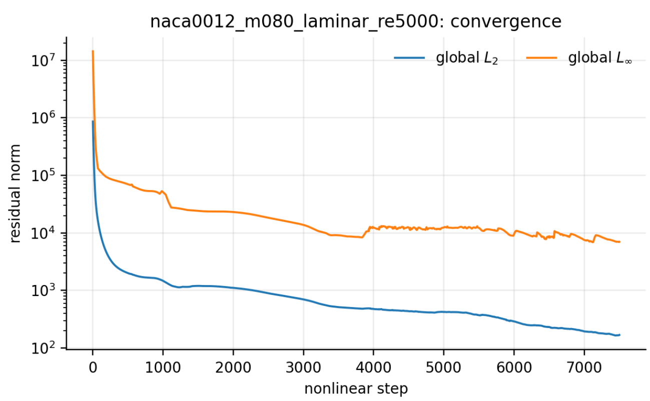
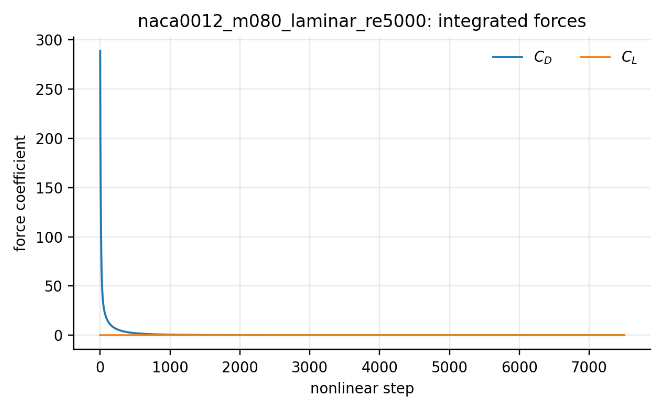


Figure 19: Wall pressure coefficient from boundary-state output for naca0012_m080_inviscid.

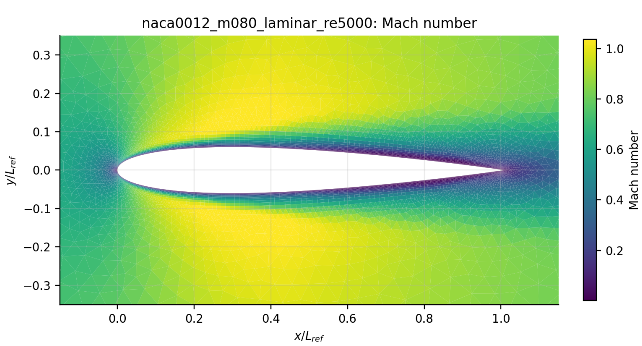


(a) Global residual norms.

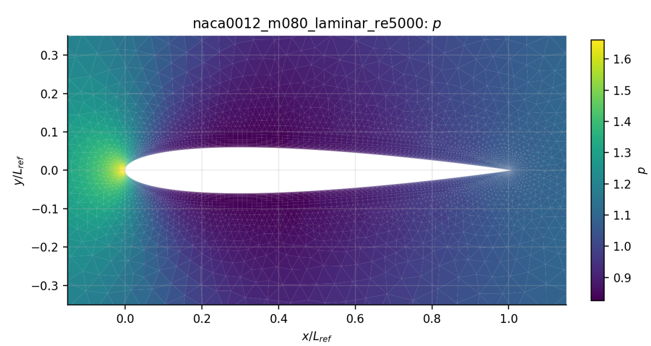


(b) Lift and drag.

Figure 20: Convergence and force histories for naca0012_m080_laminar_re5000.



(a) Mach number.



(b) Pressure.

Figure 21: Computed near-body fields for naca0012_m080_laminar_re5000; each color scale names its plotted variable.

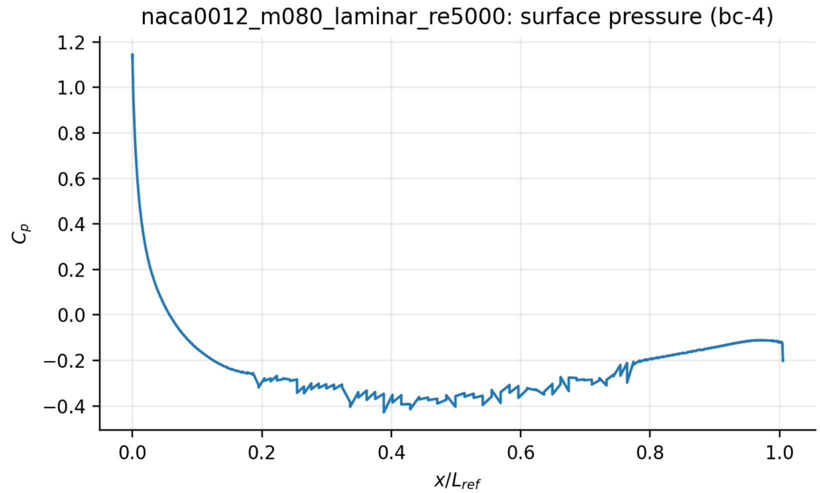


Figure 22: Wall pressure coefficient from boundary-state output for naca0012_m080_laminar_re5000.

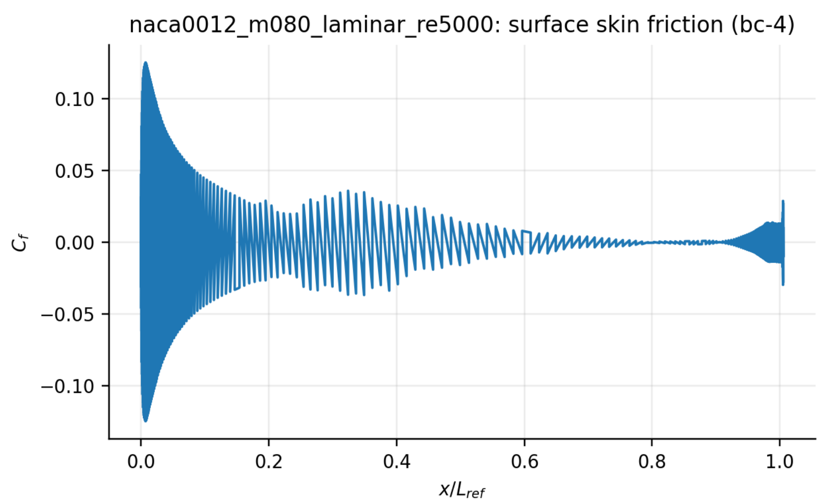
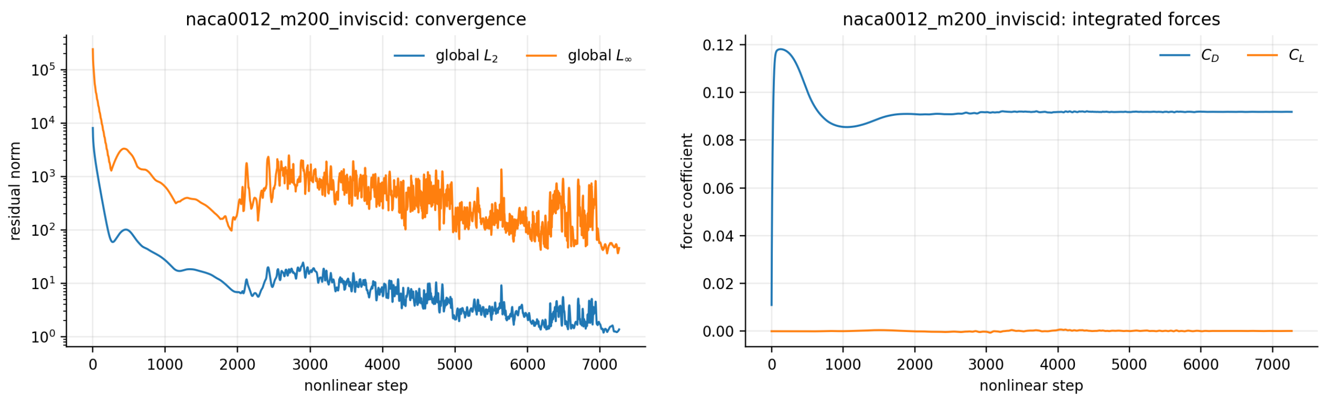


Figure 23: Tangential skin-friction coefficient for naca0012_m080_laminar_re5000.

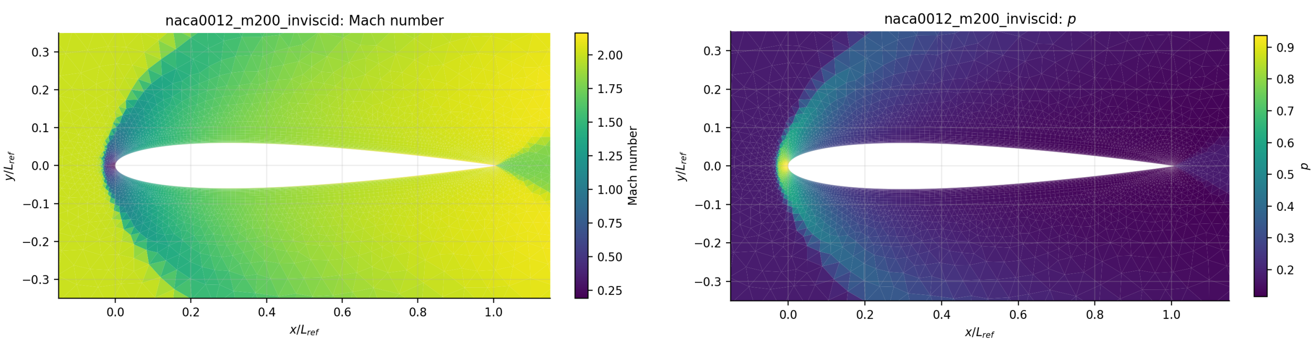
intended upper/lower symmetry at the resolution and convergence reached. The nonzero inviscid drag $C_D = 0.0918$ is dominated by Rusanov/mesh numerical diffusion (and, for the supersonic case, wave drag), not skin friction. The Mach and pressure panels expose strong supersonic compression/expansion and shock-like gradients. Figures 24–26 connect histories, loads, fields, and boundary pressure.



(a) Global residual norms.

(b) Lift and drag.

Figure 24: Convergence and force histories for naca0012_m200_inviscid.



(a) Mach number.

(b) Pressure.

Figure 25: Computed near-body fields for naca0012_m200_inviscid; each color scale names its plotted variable.

7.9 naca0012_m200_laminar_re5000

The solver classified this case as **converged** after 5382 steps. Its terminal coefficients are $C_D = 0.0391799$ and $C_L = 6.24417e - 06$. The zero-incidence lift magnitude is $|C_L| = 6.24e - 06$, consistent with the intended upper/lower symmetry at the resolution and convergence reached. Pressure and tangential viscous contributions are $C_{D,p} = 0.01242$ and $C_{D,v} = 0.02676$; the viscous share is 68.3%. The compact one-ring wall-gradient treatment makes this split more mesh-sensitive than the integrated pressure field. The Mach and pressure panels expose strong supersonic compression/expansion and shock-like gradients. Figures 27–29 connect histories, loads, fields, and boundary pressure. Figure 30 reports tangential skin friction separately from pressure drag.

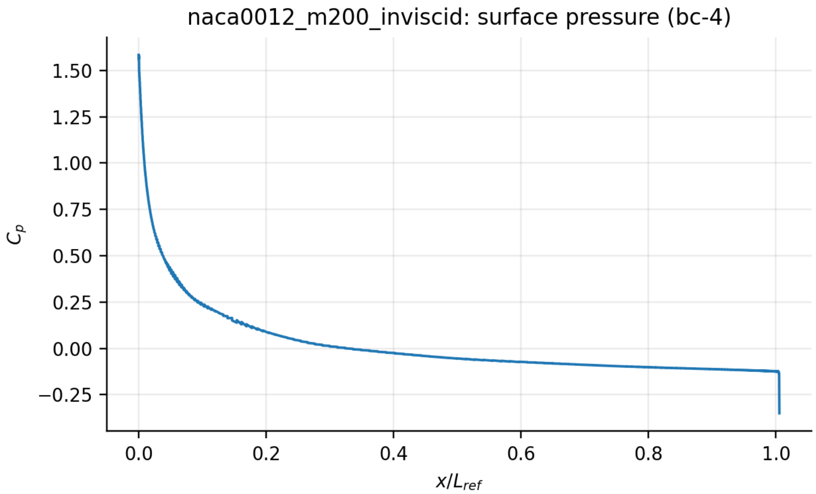
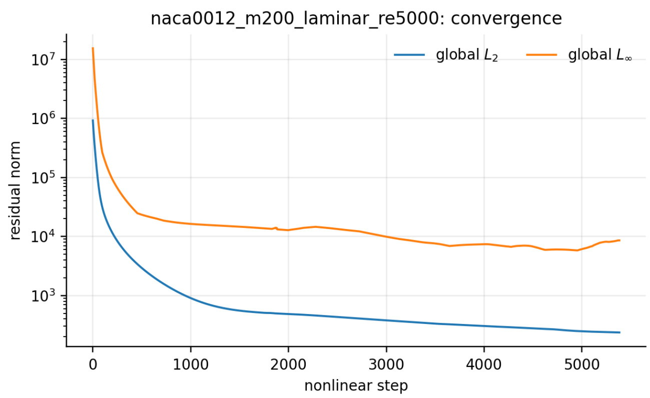
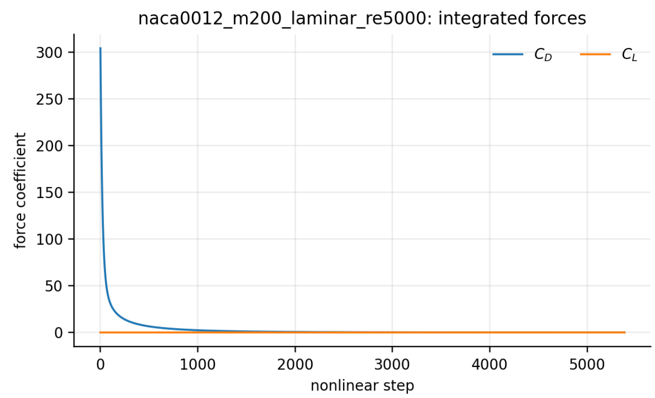


Figure 26: Wall pressure coefficient from boundary-state output for naca0012_m200_inviscid.

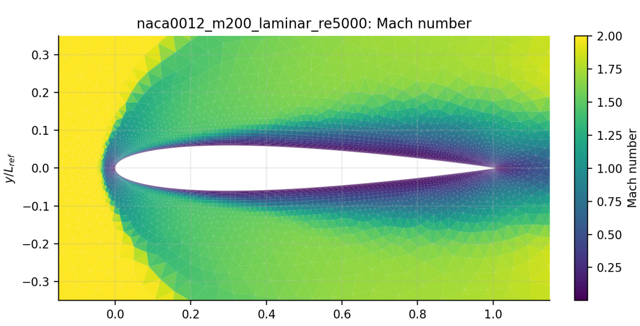


(a) Global residual norms.

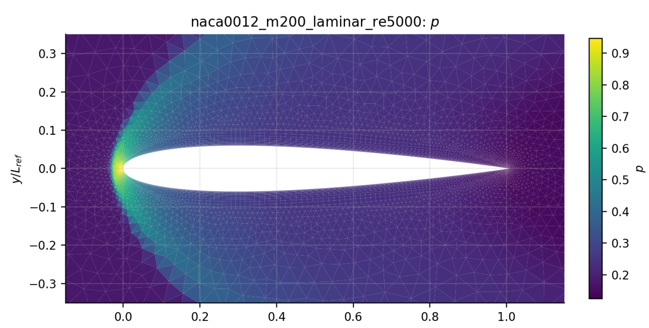


(b) Lift and drag.

Figure 27: Convergence and force histories for naca0012_m200_laminar_re5000.



(a) Mach number.



(b) Pressure.

Figure 28: Computed near-body fields for naca0012_m200_laminar_re5000; each color scale names its plotted variable.

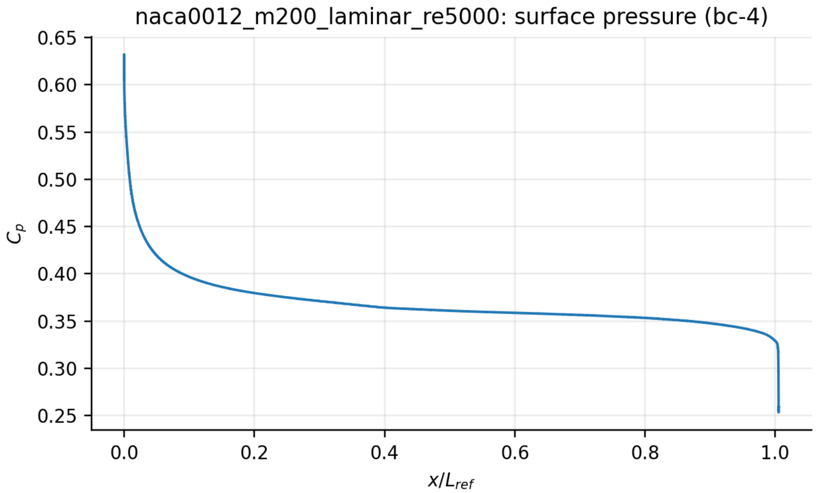


Figure 29: Wall pressure coefficient from boundary-state output for naca0012_m200_laminar_re5000.

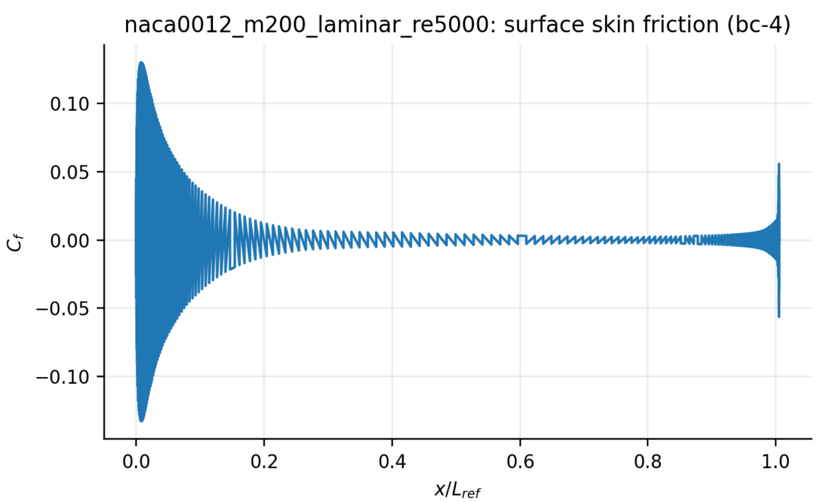


Figure 30: Tangential skin-friction coefficient for naca0012_m200_laminar_re5000.

8 Parallel validation

Production partition diagnostics and the exact rank count appear in each case package. Separate $np = 1$ and $np = 8$ consistency runs for one NACA and one cylinder case are listed in the generated parallel table when available. We compare integrated coefficients and residual reduction rather than requiring bitwise identity because METIS changes summation and limiter ordering. Final submission acceptance requires differences small relative to the physical coefficient and documents wall time, edge cut, load balance, and communication neighbors. For cylinder_m010_laminar_re20, the eight-rank run exchanged 899

Table 5: MPI rank-count consistency and timing. Residual differences are absolute differences in reduction orders.

Case	ranks	$\lvert\Delta C_D\rvert$	$\lvert\Delta C_L\rvert$	$\lvert\Delta r_{ord}\rvert$	t_1	t_8	speedup	eff.
cylinder_m010_laminar_re20	1/8	4.235e-02	2.336e-03	5.266e-02	1482.91	362.85	4.09	0.51
naca0012_m015_inviscid	1/8	3.288e-04	4.164e-03	1.213e-01	2478.60	602.60	4.11	0.51

Table 6: Eight-rank partition summary. The load-balance ratio is maximum owned cells divided by the rank mean.

Case	owned range	balance	ghosts	neighbors	sends	receives	edge cut
cylinder_m010_laminar_re20	1263–1296	1.018	899	4.5/5	899	899	476
naca0012_m015_inviscid	2565–2636	1.013	864	3.5/6	864	864	440

Table 7: Per-rank decomposition diagnostics for the representative eight-rank runs.

Case/rank	owned	ghosts	neighbors	boundary faces	sends	receives
cylinder_m010_laminar_re20/0	1270	118	5	24	118	118
cylinder_m010_laminar_re20/1	1276	110	5	24	112	110
cylinder_m010_laminar_re20/2	1263	87	3	52	87	87
cylinder_m010_laminar_re20/3	1270	133	5	0	127	133
cylinder_m010_laminar_re20/4	1267	119	5	0	120	119
cylinder_m010_laminar_re20/5	1270	121	5	0	121	121
cylinder_m010_laminar_re20/6	1273	103	3	0	106	103
cylinder_m010_laminar_re20/7	1296	108	5	20	108	108
naca0012_m015_inviscid/0	2634	123	6	51	124	123
naca0012_m015_inviscid/1	2636	96	2	73	96	96
naca0012_m015_inviscid/2	2607	112	3	61	111	112
naca0012_m015_inviscid/3	2586	114	4	58	114	114
naca0012_m015_inviscid/4	2578	92	2	64	92	92
naca0012_m015_inviscid/5	2635	101	3	65	101	101
naca0012_m015_inviscid/6	2565	169	5	32	169	169
naca0012_m015_inviscid/7	2575	57	3	80	57	57

owned-cell values per halo update across a mean of 4.5 neighbor ranks. Its measured speedup was 4.09 (51.1% efficiency); the departure from ideal scaling includes halo exchanges, global reductions, partition setup, and final gather/output overhead. For naca0012_m015_inviscid, the eight-rank run exchanged 864 owned-cell values per halo update across a mean of 3.5 neighbor ranks. Its measured speedup was 4.11 (51.4% efficiency); the departure from ideal scaling includes halo exchanges, global reductions, partition setup, and final gather/output overhead.

9 Limitations

Rusanov flux is deliberately robust but diffusive at low Mach number and across supersonic shocks. The frozen first-order block preconditioner omits exact reconstruction and viscous cross-coupling derivatives and is less efficient than a fully linearized Newton–Krylov method. Wall gradients use a compact one-ring treatment and no turbulence or transition model, so Reynolds-5000 laminar results are mesh- and model-dependent. The code currently supports triangular and quadrilateral 2-D volume cells; its topology/state interfaces leave room for 3-D faces, a general EOS, RANS closures, and species vectors, but those extensions are not claimed here. Any case-specific convergence or statistical limitation is reported beside its figures rather than hidden by a success label.